VALYON SARL · COTONOU, BENIN

Axolotl

AI that helps radiologists catch breast cancer in time.

Pre-seed · First round open · contact@valyon.co · axolotl.valyon.co

THE PROBLEM

In Africa, breast cancer kills because it is seen too late.

44%

five-year survival for breast cancer in sub-Saharan Africa [1]

32%

five-year relative survival once it has spread, vs over 99% when caught early (US data) [2]

3

radiologists per million people in Ghana, vs 100 to 120 considered adequate [3]

IT ALREADY WORKS ELSEWHERE

In Sweden, AI-supported reading finds more breast cancers.

+29%

cancers detected, with no increase in false positives [4]

−44%

screen-reading workload for radiologists [4]

MASAI: a randomised trial of 105,915 women in Sweden's national screening programme. Final results (2026): AI-supported reading misses no more cancers between screenings, with fewer aggressive ones among them. [5]

THE PRODUCT

A second reader for every mammogram.

01

Import

DICOM files from the hospital PACS or a single workstation.

02

Analyse

The model outlines and measures suspicious areas, and ranks studies by urgency.

03

Decide

The radiologist confirms, corrects or dismisses every flag.

Axolotl flags. The radiologist decides.

OUR METHOD

Proven elsewhere. Adapted and validated here.

MASAI used one country, one mammography system and one AI product. In Benin's public hospitals, breast imaging remains poorly accessible, and a national screening programme is still to be created. [5] [6]

1. Evaluate

Test MedGemma as-is on public mammograms: our baseline.

2. Adapt

Fine-tune it with LoRA, first on public, then on regional mammograms.

3. Re-evaluate

Measure the gain on local studies with partner radiologists, before any clinical use.

MARKET AND BUSINESS MODEL

Three customers, one product.

Customer	Their need	How they pay
Private clinics and imaging centres	Faster reports, safer reading	Monthly subscription per site
Public hospitals and university hospitals	Long queues, too few specialists	Annual licence, with deployment and training
Screening programmes: ministries, NGOs, Octobre Rose campaigns	Reading peaks over a few weeks	Fee per mammogram analysed

Benin first, then the eight UEMOA countries: one currency, one regional framework.

COMPUTE AND STACK

About $4k of GPU time in year one.

Phase	AWS EC2 instance	GPU hours	Cost
1. Baseline evaluation (month 1)	g6.xlarge · 1× NVIDIA L4	~100 h	~$80
2. Fine-tuning (months 2–4)	g6e.xlarge · 1× L40S, plus L4	~310 h / mo	~$515 / mo
27B model trial (one-off)	g6e.12xlarge · 4× L40S	~40 h	~$420
3. Pilot inference (months 5–12)	g6.xlarge, business hours	~260 h / mo	~$250 / mo

Stack: MedGemma · PyTorch · MONAI Label · NVIDIA FLARE · TensorRT · Amazon EC2 · Amazon S3 with AWS KMS

Target: ~$10k in credits, with storage, data transfer and a 2× margin for experiments

07 · On-demand, us-east-1: g6.xlarge $0.80/h · g6e.xlarge $1.86/h · g6e.12xlarge $10.49/h [7]

DATA PROTECTION

Mammograms stay under each hospital’s control.

Federated learning

With NVIDIA FLARE, the model will train inside each hospital. The images stay there.

Benin’s law

Book V of the Code du numérique applies. Our APDP authorisation request is in progress.

Never resold

Encrypted with AWS KMS and used only for analysis. Each facility keeps control of its images.

ROADMAP

From prototype to regional deployment.

In progress

Prototype

Evaluate MedGemma on public mammograms, fine-tune it, and build the radiologist's review interface.

In preparation

Pilot in Benin

In parallel: partner hospitals, APDP authorisation and ethics approval, then the first real mammograms.

Next

Validation

Performance study with partner radiologists, and regulatory steps.

Later

UEMOA

First commercial contracts, then neighbouring countries.

TEAM

The founders.

Jean-Baptiste Carmel Togninou Adikin

Co-founder and co-manager

Research at IRISA and Inria; systems security.

Yelian Constantin Zounmevo

Co-founder and co-manager

[One line of background]

Hiring next: AI engineers · a medical lead · a deployment lead

WHAT WE NEED

Help us bring a proven approach here.

Compute

About $10k of GPU credits for evaluation, fine-tuning and pilot inference.

A starter dataset

Public mammograms first, then local studies collected with APDP authorisation and ethics approval.

Partner radiologists

To annotate mammograms and lead the validation.

First funding

To run the clinical pilot in Benin.

Sources

[1] “Prognostic factors of breast cancer and survival in Sub-Saharan Africa: a systematic review and meta-analysis”, Bulletin du Cancer, 2025, 112: 1024–1034.

[2] American Cancer Society, “Survival Rates for Breast Cancer”, SEER data, women diagnosed between 2015 and 2021.

[3] Sarkodie B. D. et al., “Density and regional distribution of radiologists in a low-income country: the Ghana situation”, Chinese Journal of Academic Radiology, 2023, 6(4): 188–195.

[4] Hernström V. et al., “Screening performance and characteristics of breast cancer detected in the MASAI trial”, The Lancet Digital Health, 2025, 7(3): 175–183. doi:10.1016/S2589-7500(24)00267-X

[5] Gommers J. et al., “Interval cancer, sensitivity, and specificity comparing AI-supported mammography screening with standard double reading without AI in the MASAI study”, The Lancet, 2026, 407(10527): 505–514. doi:10.1016/S0140-6736(25)02464-X

[6] Yekpe Ahouansou P. et al., “Accessibilité à l’imagerie dans la prise en charge des cancers du sein en milieu hospitalier au Bénin”, Journal Africain d’Imagerie Médicale, 2022, 14(1): 8–13.

[7] Amazon EC2 on-demand pricing, us-east-1 region, g6 and g6e instance families, 2026.

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